

MAHADEV

Final-Year B.E. Electrical Engineering Student Power Systems | Renewable Energy | Solar PV & BESS | SCADA | MATLAB/Simulink | Embedded & IoT

📍 Address: Hyderabad, Sindh, Pakistan 📞 Phone: +923327838365 ✉ Email: mahadevkhokhar05@gmail.com
🔗 LinkedIn: [linkedin.com/in/mahadevkumar](https://www.linkedin.com/in/mahadevkumar) 📱 portfolio-mahadev.vercel.app/

PROFESSIONAL PROFILE

Final-year B.E. Electrical Engineering student at Mehran University of Engineering and Technology (MUET), Jamshoro, with hands-on experience in power systems, renewable energy, Solar PV & BESS, SCADA/instrumentation, embedded systems and IoT. Completed five industrial internships across Abtex International, SSGC, HESCO—including the 132 kV Qasimabad Grid Station—and JPCL, gaining exposure to PV-grid-battery power flow, LT distribution, substations, protection and switching, thermal generation, SCADA/RTUs and industrial electrical systems. Developed MATLAB/Simulink models and ESP32/Arduino prototypes covering circuit design, firmware, fabrication, testing and technical documentation. Currently completing a Final Year Project on industrial exhaust-air energy recovery using a hybrid VAWT-PMSG system.

EDUCATION

B.E Electrical Engineering	2022 – 2026
Mehran University of Engineering & Technology (MUET), Jamshoro CGPA: 3.00 / 4.00	
Intermediate (Pre Engineering)	2022
Govt. H/Sec School Nabisar Sahar	
Matriculation (Science)	2020
Govt. H/Sec School Nabisar Sahar	

INDUSTRIAL EXPERIENCE

Electrical Engineering Intern — Solar PV & BESS

Abtex International (Pvt.) Ltd. | Hyderabad, Sindh (1 Month) 1 Jul 2026 – 31 Jul 2026

- Gained practical exposure to Solar PV generation, inverter operation, BESS/PCS, LT distribution and grid import/export monitoring.
- Reviewed PV-grid-battery power flow, battery charging/discharging and industrial plant-load behaviour.
- Analysed solar utilisation and energy-efficiency opportunities and prepared weekly and final technical reports.

Industrial Intern — Electrical Engineering

Sui Southern Gas Company Limited (SSGC) | Hyderabad, Sindh (6 Weeks) 29 Jun 2026 – 10 Aug 2026

- Gained exposure to SCADA and instrumentation, including SCADAPack RTUs, HMI concepts, radio telemetry and industrial process instrumentation.
- Observed generator/powerhouse systems, three-phase distribution, switchgear, HVAC equipment, VFDs, pumps and motor-control applications.
- Studied microwave communication systems, gas-meter testing and industrial utility operations and prepared technical internship documentation.

Power Systems Intern — 132 kV Qasimabad Grid Station

Hyderabad Electric Supply Company (HESCO) | Qasimabad, Hyderabad (4 Weeks) July 2025 – Aug 2025

- Observed 132/11 kV grid-station operation, protection, switching and maintenance practices.
- Studied power transformers, CTs, PTs, SF₆/VCBs, protection relays, isolators and busbars.
- Reviewed single-line diagrams, feeder operation, load management and high-voltage safety procedures.

Electrical Engineering Intern — Thermal Power Generation

Jamshoro Power Company Limited (JPCL), GENCO-I | Jamshoro (1 Month)

Jun 2024 – Jul 2024

- Observed thermal power generation, steam turbines, generators, transformers, electrical panels and control-room monitoring.
- Studied instrumentation, industrial safety, plant maintenance and troubleshooting practices.

Electrical Engineering Intern

Hyderabad Electric Supply Company (HESCO) | Hyderabad (2 Weeks)

Dec 2023 – Jan 2024

- Observed distribution system operation, feeder planning, and grid safety practices.
- Studied conductor selection, protection coordination, and distribution management.
- Gained field exposure in Chief Engineer P&E and GSO HESCO Hyderabad offices.

PROJECTS

MY FINAL YEAR PROJECT (ONGOING)

Industrial Exhaust-Air Energy Recovery Using a Hybrid VAWT–PMSG System

- Evaluating pressure-loss-constrained energy recovery from industrial exhaust airflow using a hybrid vertical-axis wind turbine and PMSG configuration.
- Developing MATLAB/Simulink models covering PMSG generation, rectification, DC–DC conversion, MPPT control and battery/storage integration.
- Preparing feasibility analysis, system diagrams and technical research documentation.

INDEPENDENT CONTRACT PROTOTYPE

IoT-Based Smart Energy Management with Theft Detection

Dec 2025

ESP32 | Solar/Grid | ATS | IoT Dashboard | Completed

- Designed a dual-source energy system using solar/inverter and WAPDA grid supply.
- Integrated automatic transfer switching for uninterrupted source changeover.
- Monitored solar voltage, battery voltage, grid voltage, load current, and power.
- Developed theft detection using ACS712 sensing and abnormal current monitoring.
- Built a Wi-Fi-based dashboard for live source, alarm, and system-status monitoring.
- Integrated ESP32, ZMPT101B, ACS712, relays, LCD, inverter, battery, and solar panel.

Design and Fabrication of an Automatic Solar Panel Cleaning System

Jan 2026

Arduino UNO | BTS7960 (IBT-2) | RTC | Bluetooth | Relay Control | Completed

- Fabricated a solar panel cleaning prototype using chain drive and wiper mechanism.
- Controlled bidirectional DC motor movement using Arduino UNO and BTS7960 driver.
- Developed Bluetooth manual control using HC-05 and scheduled control using DS3231 RTC.
- Integrated relay-based water pump control for synchronized cleaning operation.
- Used limit switches for safe automatic stopping at both ends of the panel.
- Tested wiring, motor movement, pump operation, and cleaning mechanism reliability.

IoT-Based LPG and Natural Gas Leakage Detection System with Automatic Servo-Actuated Flow Control

2026

ESP32 | MQ-2 | MQ-5 | Flame Detection | Servo Control | IoT Dashboard | Completed

- Developed an ESP32-based system for LPG, natural-gas, and flame detection with automatic safety control.
- Integrated MQ-2, MQ-5, flame sensor, servo valve, exhaust fan, buzzer, LEDs, LCD, RTC, and microSD logging.
- Implemented SAFE, WARNING, GAS LEAK, and FIRE modes with automatic valve shut-off and alarm response.
- Developed a local Wi-Fi dashboard for real-time monitoring and authorised manual control.
- Designed the EdrawMax circuit diagram and completed firmware, soldering, calibration, testing, and hardware integration.

Design and Fabrication of Smart Energy Meter with Priority Load Control

ESP32 | PZEM-004T | CT | Relay Control | Load Prioritization | Prepaid Logic

Ongoing

- Developed a smart energy meter concept with automatic priority load control.

- Integrated voltage, current, power, energy, frequency, and power factor monitoring.
- Designed load-shedding logic to disconnect low-priority loads during overload.
- Used ESP32, PZEM-004T, CT, relay module, OLED/LCD, buzzer, and load circuit.
- Prepared block diagram, circuit diagram, and implementation workflow.

Smart Monitoring of Transmission Lines for Enhancing Power Grid Reliability

MATLAB/Simulink | Power Systems | Fault Detection | Sag | Temperature | Vibration | Completed

- Developed a MATLAB/Simulink model for transmission-line condition monitoring.
- Simulated voltage, current, temperature, sag, vibration, overload, and fault cases.
- Designed threshold alarm logic for overcurrent, undervoltage, and overtemperature.
- Used RMS, per-unit voltage, loading index, thermal response, and sag estimation.

SolarSense AI — AI-Assisted Solar Monitoring & Automatic Cleaning System

ESP32 | Solar PV | Computer Vision | Web Dashboard | Completed

- Developed an ESP32-based solar-monitoring and automatic-cleaning prototype integrated with AI-assisted panel-condition assessment.
- Integrated environmental sensing, motor/pump control, dashboard monitoring and computer-vision detection for dust severity, surface cracks and bird droppings.
- Contributed to component selection, hardware fabrication, firmware, AI integration, testing and technical documentation.

KitchenGuard — ESP32 Smart Kitchen Safety System

ESP32 | Sensors | IoT | Automation | IEEE SSCS Arduino Contest 2026 | Completed

- Developed an ESP32-based safety prototype integrating environmental sensing, presence/cookware detection, OLED display, alerts, ventilation, servo control and web monitoring.
- Performed circuit integration, ESP32 programming, sensor calibration, actuator control, testing and troubleshooting.

CERTIFICATIONS

• IEEE SSCS Arduino Contest — Micro-Controller Proficiency Program	2026
• IEEE SSCS Arduino Contest — Certificate of Participation	2026
• Arduino Innovation Hackathon — MUET / IEEE SSCS	2026
• Introduction to AI — Google via Coursera	2026
• Clean Energy and Climate Resilient Housing Workshop – RDF	2026
• MATLAB Onramp – MathWorks	2025
• IoT and Startup Culture Workshop – QS ImpACT Pakistan and Indus Techetronics	2024
• Condition-Based Maintenance of Equipment in Power Sector – IEEE PES	2024
• Geothermal Energy Extraction and Infrastructure Planning – Technical Session	2024
• IEEE Day 2024: Research and Innovation with IoT & AI	2024
• Unlocking Scholarship Opportunities for Undergraduates – IEEE PES MUET	2024

RESEARCH AND ACADEMIC PRESENTATION

Poster Presenter

8th International Conference on Energy, Environment and Sustainable Development (EESD 2026)

MUET Jamshoro | 25–26 March 2026

- Presented a poster at the international conference organized by the Energy & Environmental Engineering Research Group (EEERG), MUET Jamshoro.
- Focused on energy, environment, sustainable development, and applied engineering research.

PROFESSIONAL MEMBERSHIPS

IEEE (Institute of Electrical and Electronics Engineers)	2024
• IEEE Student Member (Karachi Section)	
• IEEE Power & Energy Society (PES)	

- IEEE Circuits & Systems Society
 - IEEE Industry Applications Society
-

TECHNICAL SKILLS

Power & Energy Systems

- Power systems, substations, protection and switching, transmission and distribution, Solar PV, BESS/PCS, grid-connected energy systems, energy management, MPPT, power electronics.

Simulation & Engineering Software

- MATLAB, Simulink, ETAP, Proteus, Multisim, EdrawMax.

SCADA & Industrial Systems

- SCADA/HMI concepts, RTUs, radio telemetry, instrumentation, VFDs, generators, LT distribution and industrial electrical systems.

Embedded Systems & IoT

- ESP32, Arduino UNO, Embedded C/C++, sensor interfacing, relay control, motor control, local web dashboards and data logging.

Hardware & Testing

- Circuit prototyping, Veroboard fabrication, soldering, wiring, calibration, testing, debugging and troubleshooting.

Engineering Documentation

- Single-line diagrams, block diagrams, circuit diagrams, technical reports, MS Word, Excel and PowerPoint.
-

LANGUAGES

- English – Professional Working Proficiency
- Urdu – Native / Fluent
- Sindhi – Native / Fluent